

Weight Initialization Techniques

Key insights for improving neural-network training

Things not to do

1. Zero initialization
2. Non-zero constant initialization
3. Random initialization with small weights
4. Random initialization with large weights

What can be done

- **Xavier / Glorot initialization** - commonly used with sigmoid or tanh activations; it helps keep activation and gradient variance stable across layers.
- **He initialization** - designed especially for ReLU and ReLU-like activations; it helps reduce vanishing or exploding signals in deep networks.

***Quick rule:** Xavier/Glorot is a strong default for tanh/sigmoid networks, while He initialization is generally preferred for ReLU-based networks.*